

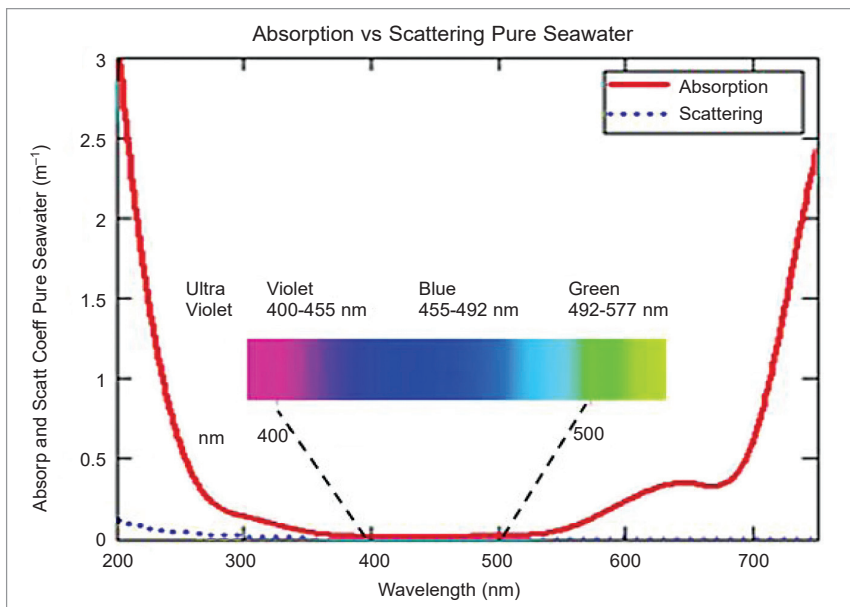


Fig. 2: Absorption and scattering for different wavelengths (Source: <https://iopscience.iop.org>)

ter, clear ocean water, coastal ocean water, and turbid harbour water.

Receiver. The receiver module consists of a lens to collect incident optical signals from the transmitter, an avalanche photo-diode, for capturing the optical signals and converting them into photocurrent, a trans-impedance amplifier to convert photocurrent into voltage values, an analogue to digital converter to convert analogue voltage into digital signals for data detection, and a data detection module to retrieve the transmitted data corrupted by the channel noise and detector imperfections.

Factors affecting underwater optical communication

Some major factors that affect the underwater optical propagation are:

Water type. We encounter various types of water, each with its unique characteristics. These water types include pure seawater, transparent ocean water, coastal ocean water, and murky harbour water. Each of these water types exhibits distinct physical properties, including differences in

temperature, impurity levels, and salinity levels.

Absorption and scattering. The attenuation of the optical signal in the underwater channel is mostly contributed to by absorption and scattering in the water. The beam extinction coefficient, c , for the attenuation is given as $c(\lambda) = a(\lambda) + b(\lambda)$.

where a is the absorption coefficient and b is the scattering coefficient. Both coefficients are dependent on the wavelength being used for communication.

Particles in water. Various dissolved salts, ground quartz, clay minerals, metal oxides, coloured dissolved organic matter and other organic matter such as viruses, colloids, bacteria, phytoplankton, zooplankton, and organic detritus also affect the optical signal propagation in the water.

Turbulence. It can create signal attenuation, beam spread, signal distortion, modal noise, and signal disruption in underwater optical communication.

Pointing and alignment. Pure seawater requires tight pointing requirements as the optical beam is likely to follow a collimated

path. On the other hand, in turbid harbour, scattering is the significant contributor causing the optical beam to spread in different directions. A misalignment of three degrees reduces the detector response by a factor of hundred.

Background noise. Scattered light collected by the receiver, background noise from the Sun, and diffused extended background noise affect the UWOC.

Wavelength used in UWOC system

Fig. 2 shows the absorption and scattering of various wavelengths in pure seawater. The bandwidth used in UWOC is typically 400-500nm due to their minimum attenuation (low absorption and low scattering), which is a key driver of high data rate and longer distance. Most of the undergoing projects are working at this band for the UWOC system.

Advantages of UWOC

The UWOC has several advantages over traditional radio and acoustic communication systems. Some key advantages are:

High data rates. One of the significant advantages of underwater optical communication is its high data transfer rates. It can transmit data at rates several times faster than traditional acoustic communication (up to Gbps data rates over a distance of tens of metres).

Low latency. Optical signals travel at the speed of light, which results in lower latency compared to acoustic signals.

Low interference. Unlike acoustic communication, optical communication is not affected by underwater noise from marine life or other sources.

Secure. Laser signals are highly directional, making it difficult for eavesdroppers to intercept the communication.